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Structural and thermochemical study of Na_2O–ZnO–P_2O_5 glasses

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ABSTRACT

Glasses of the $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$($3 \le O/P \le 3.49$) and $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ ($O/P=3$) ($0 \le x \le 33$ mol%) compositions were prepared using the conventional melt quenching technique. The increase of density and glass transition temperature in both series is related to the reticulation of phosphate chains. For the first series of glasses, Fourier-transformed infrared (FTIR), Raman and ^{31}P solid state magic angle spinning nuclear magnetic resonance (MAS-NMR) spectroscopy revealed the decrease of Q^2 tetrahedral sites and the increase of phosphate dimers (Q^1), indicating the shortening of phosphate chains. For the second series, FTIR and Raman spectroscopy show only the presence of Q^2 tetrahedral sites.
Dissolution of these series in 4.5% weight of H_3PO_4 solution has been followed calorimetrically and showed that the dissolution of the first series is endothermic for the low ZnO content and becomes exothermic when x rises. This behavior is correlated to the structural modification. For the second series, the dissolution phenomenon is endothermic confirming the presence of the same structure over the whole composition range.

1. Introduction

Alkali, alkaline earth and some transition metal oxides form glasses when melted with P_2O_5. Due to their lower melting and transition temperatures, ultraviolet transmission, high thermal expansion and optical characteristics, phosphate glasses have been the subject of many investigations over the years in order to use them in various applications such as the manufacturing of glass-polymer composites [1–3], glass to metal seals [2–7], optical waveguides [1,2,6], biomaterials [6–8], solid state laser sources [1–4,6,9,10], conducting materials [7,8,11,12], amorphous semiconductors [9,12,13], optical fibers for communication devices [10], and nuclear waste immobilization matrices [6,7].

During the recent years, transition metal oxides have attracted much attention because of their electric, magnetic and optical properties. When glasses are doped with transition metal ions, they take a place in the new laser and luminescence materials [6,7,10]. However, their relatively poor chemical durability makes them unsuitable for many applications [5,14,15]. On the other hand, it's known that transition metal oxides can play a dual role as a network former and modifier on the glass structure. They allow the formation of the ionic cross linking such as P–O–Zn which improves their chemical durability [5,14,16,17].

The conventional notation for phosphate groups adopted in this paper is Q^n, where n ($n=0$–3) is the number of bridging oxygen per PO_4 tetrahedron [2,5,18,19]. Recently, many investigators used the O/P ratio in order to classify the distribution of phosphate groups in the vitreous network. For $2.5 < O/P < 3$, ultraphosphate groups are dominant where the structure is described by a three dimensional network [19].

Metaphosphate glasses are obtained for O/P equal to 3. They have a network formed by chains and rings which are linked by more ionic bonds between the various metal cations and the non-bridging oxygens. Polyphosphate glasses are obtained for $3 \le O/P \le 3.5$. They have a network based entirely on chains formed by phosphate tetrahedron which shared two bridging oxygens. The average chain length becomes progressively shorter as the O/P ratio increases [19].

At $O/P=3.5$, the structure is dominated by phosphate dimers such as pyrophosphate groups in which two PO_4 tetrahedra are linked by a common bridging oxygen [19].

Glasses for which $3.5 < O/P < 4$ contain isolated tetrahedron such as orthophosphate units (PO_4^{3-}) [19].

In addition to that, in phosphate glasses, the spin $-1/2$ nucleus of ^{31}P is a primary source for structural information by NMR measurements.

Their glass transition temperature (T_g) depends on the composition. The increase in T_g value indicates a definite modification of the glass network suggesting that the effect of ZnO is to decrease the condensation of metaphosphate chains [17,18].

Glasses of the $(100-x)NaPO_3$–$xZnO$ ($0 \le x \le 33.3$ mol%) were reported previously by Montagne et al. [17,18]. Vitreous structure were

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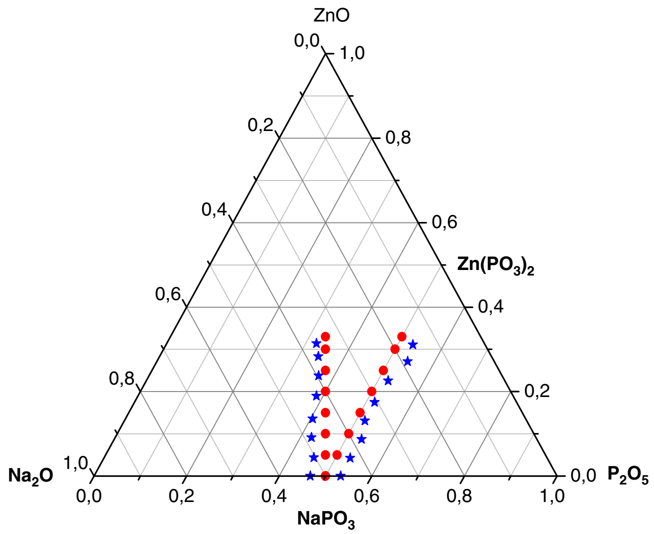

Fig. 1. Analyzed (★) and nominal (●) glass compositions for both series of glasses: $(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ and $(50 - x)Na_2O$–$xZnO$–$50P_2O_5$ ($0 \le x \le 33$ mol%) phosphate glasses.

performed by solid and liquid nuclear magnetic resonance spectroscopy (^{31}PNMR), refractive index and density measurements, visible spectroscopy, activation energy, glass durability, ICP analysis, differential scanning calorimetry (DSC) and dynamic viscosity [17,18]. However a few studies were reported on the calorimetric dissolution of phosphate glasses [20,21].

This work deals with the synthesis, chemical characterization (density, elementary analysis-ICP, DSC-DTA), structural characterization (^{31}P MAS-NMR, (FTIR) and Raman spectroscopies) and calorimetric study of $(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ and $(50 - x)Na_2O$–$xZnO$–$50P_2O_5$ ($0 \le x \le 33$ mol%) that belong to the $NaPO_3$–ZnO and $NaPO_3$–$Zn(PO_3)_2$ lines as shown in Fig. 1.

2. Experimental

2.1. Glass preparation

$(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ and $(50 - x)Na_2O$–$xZnO$–$50P_2O_5$ glasses with x ranging from 0 to 33% mol, were prepared by melting the homogenized mixture of: NaH_2PO_4 (Sigma-Aldrich), $(NH_4)_2HPO_4$(Fluka) and ZnO (Sigma-Aldrich) (99.7% purity) in the suitable proportions.

Batches of about 10 g were melted in a platinum crucible at 200 °C for 30 min, then preheated at temperatures between 400 and 600 °C in order to eliminate H_2O and NH_3. The temperature was then progressively raised to 700 °C–850 °C then held constant for 30 min. The batch was finally quenched to room temperature under air in order to produce vitreous structure which is confirmed by X-ray diffraction. All the products were annealed at about 20 °C bellow their glass transition temperature for 2 h, followed by slow cooling (1 °C/min) in order to eliminate internal tensions and get more homogenized samples.

2.2. ICP analysis

The content of phosphorus, zinc and sodium was analyzed by means of inductively coupled plasma atomic emission spectroscopy (Jobin Yvon Ultra C). The nominal and analytical glass compositions are reported in Table 1 $(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ and $(50 - x)Na_2O$–$xZnO$–$50P_2O_5$ glasses will be labeled by reference to their molar ZnO content x ranging from 0 to 33 mol%.

2.3. Measurements of density

Glass density measurements were made using the standard Archimedes' method with diethyl orthophthalate as the immersion fluid. The relative error of these measurements is ±3%. The molar volume was calculated from the density ($V = M/\rho$) and the molar weight M.

2.4. DSC investigations

The glass transition temperatures were determined on 40–50 mg of samples using DSC-ATD Netzsch 404 PC with a 10 °C/min heating rate (accuracy ±5 °C).

2.5. FTIR, Raman and NMR spectroscopies

FTIR spectra were recorded with an FTIR spectrometer (Bruker) in the frequency range of 40–7500 cm^{-1} at room temperature. The samples were prepared by grinding about 9 mg of glass powder with 300 mg spectroscopic grade dried KBr. All measurements were at 4 cm^{-1} resolution.

The Raman spectra were recorded on powder of glasses by means of a Labram HR 800 micro Raman model operating in the 50–4000 cm^{-1} range at room temperature equipped with an internal He–Ne laser source ($\lambda = 488$ nm).

The ^{31}P MAS-NMR spectra were collected at 121.456 MHZ (magnetic field 7.1 T) on a Bruker ultrashield 300 spectrometer and at a sample spin rate of 8 kHz. All chemical shifts are expressed in ppm relative to an 85% aqueous phosphoric acid. In general chemical shifts were independent on experimental parameters. The chemical shift resolution is ±0.2 ppm.

2.6. Calorimetric dissolution

The dissolution of phosphate glasses were performed using the C80 (SETARAM) calorimeter operating at 25 °C. This device is provided with two identical cells: the reference and the measuring cell. The last one contains the solid to be dissolved which is tightly separated from the solvent by a movable cover. Whereas the reference cell was provided with only the solvent.

Both of the cells are surrounded by high performance thermopiles for the detection of heat flow resulting from the dissolution of about 20 mg solid in 4.5 ml of a 4.5% weight phosphoric acid solution. The

Table 1
Analyzed and nominal glass compositions for both series of glasses: $(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ and $(50 - x)(Na_2O)$–$xZnO$–$50P_2O_5$ ($0 \le x \le 33$ mol%) phosphate glasses.

ZnO nominal/analyzed	Na_2O nominal/analyzed	P_2O_5 nominal/analyzed	ZnO nominal/analyzed	Na_2O nominal/analyzed	P_2O_5 nominal/analyzed
0/0	50/46.7	50/53.3	0/0	50/46.7	50/53.3
5/4.4	47.5/45.3	47.5/50.3	5/4.3	45/42.6	50/53.2
10/9.1	45/42.4	45/48.5	10/8.7	40/37.8	50/53.4
15/13.5	42.5/40.4	42.5/46	15/13.1	35/35	50/52.0
20/18.9	40/38.6	40/42.5	20/17.5	30/30.6	50/51.8
25/23.8	37.5/34.3	37.5/39.7	25/22.7	25/25.2	50/52.4
30/28.3	35/34.3	35/37.4	30/27.1	20/18.7	50/54.2
33/31.4	33.5/32.4	33.5/36.3	33/31.1	17/15.5	50/53.3

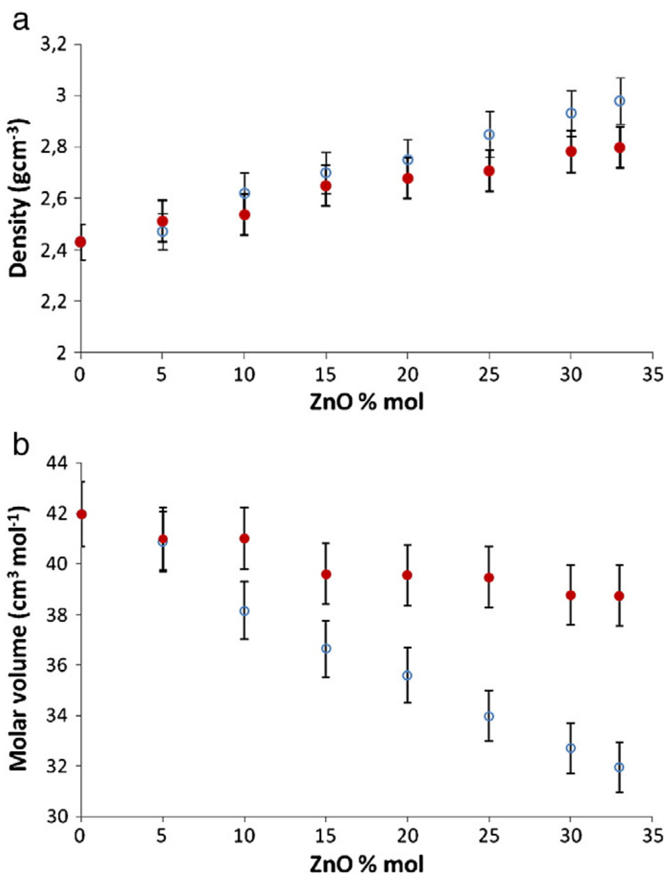

Fig. 2. a. Density dependence of $(50-x/2)Na_2O–xZnO–(50-x/2)P_2O_5$ (○)and $(50-x)Na_2O–xZnO–50P_2O_5$ ($0 \le x \le 33$ mol%) (●) ($0 \le x \le 33$ mol%) over ZnO content. b. Molar volume evolution of $(50-x/2)Na_2O–xZnO–(50-x/2)P_2O_5$ (○)and $(50-x)Na_2O–xZnO–50P_2O_5$ ($0 \le x \le 33$ mol%) (●) ($0 \le x \le 33$ mol%) over ZnO content.

dissolution enthalpy was determined by integrating the surface of the recorded signal. The fiability of the calorimeter was verified by the key dissolution reaction [22,23].

3. Results and discussion

3.1. Density and molar volume

The variations of density and molar volume of the first series $(50-x/2)Na_2O–xZnO–(50-x/2)P_2O_5$ where $3 \le O/P \le 3.49$ and the second series $(50-x)Na_2O–xZnO–50P_2O_5$ with $O/P = 3$ phosphate glasses versus the %mol ZnO ($0 \le x \le 33$ mol%) are presented in Fig. 2(a, b).

The density of both series of glasses (Fig. 2a) increased gradually with the introduction of ZnO content. The increase in density indicates that the ZnO oxide reticulates the vitreous network, because P–O–Zn bond are more ionic than P–O–P bond, allowing the compactness of the structure [9,14].

Fig. 2b shows that the molar volumes, $V_m = \Sigma x_i M_i/\rho$ (where M_i is the molar weight of the component i) decrease monotonically with the increase of ZnO content for the first series of glasses (x = 0 mol%, $V_m = 42 cm^3 mol^{-1}$; x = 33 mol%, $V_m = 32 cm^3 mol^{-1}$) indicating the reticulation of glass network which is tightly related to the higher electric field strength noted $\Delta F = z/r^2$ (with z is the valence cation and r is the ionic radius of the corresponding atom) of Zn^{2+} compared to Na^+ [9,18]. This behavior is characteristic of ZnO effect that decreases the average chain length of metaphosphate glasses resulting from the following reaction: [17,18].

$$P—O—P + ZnO \rightarrow (2PO)Zn.$$

The molar volume V_m decreases linearly with increasing ZnO content, confirming that ZnO reticulates the vitreous network [17,18].].

For the second series of glasses, the variation of V_m is less important (x = 0 mol%, $V_m = 42 cm^3 mol^{-1}$; x = 33 mol%, $V_m = 38.74 cm^3 mol^{-1}$) than in the first one. This behavior could be due to the conservation of the structure formed by metaphosphate chains. The molar volume variation may be due to the substitution of a big ion Na^+ (r = 1 Å) by a small one Zn^{2+} (r = 0.74 Å) and also by replacing Na–O bond by the more covalent bond Zn–O.

3.2. DSC study

The variation of glass transition temperature is shown in Fig. 4. T_g values increase with ZnO content for both series of glasses as mentioned in Table 2.

Further, DSC curves were presented in Fig. 3 (a, b). It seems that for $40Na_2O–10ZnO–50P_2O_5$ and $17Na_2O–33ZnO–50P_2O_5$ glass compositions, DSC curves exhibit two peaks of crystallization. This result can be due to the existence of more than one amorphous phase as mentioned in Fig. 3b.

The glass transition temperature increases from 280 °C to 314 °C for $(50-x/2)Na_2O–xZnO–(50-x/2)P_2O_5$ and from 280 °C to 311 °C for $(50-x)Na_2O–xZnO–50P_2O_5$.

We noticed the same variation of glass transition temperature for both series of glasses. Thus, the increase in T_g reflects an increase of the network cross-link strength of the structure as Zn^{2+} ions are

Table 2
Glass composition, glass transition temperature T_g, T_c, ΔT, density ρ, molar volume V_m, of $(50-x/2)Na_2O–xZnO–(50-x/2)P_2O_5$ and $(50-x)Na_2O–xZnO–50P_2O_5$ ($0 \le x \le 33$ mol%).phosphate glasses. ($\Delta T = T_c - T_g$).

	X	T_g(°C)	T_c(°C)	ΔT(°C)	Density (g/cm^3)	Vm (cm^3/mol)
$(50-x/2)Na_2O–xZnO–(50-x/2)P_2O_5$	0	280 ± 5	290 ± 5	10	2.43 ± 0.07	42.00 ± 1.30
	5	–	–	–	2.47 ± 0.07	41.00 ± 1.23
	10	285 ± 5	371 ± 5	86	2.62 ± 0.08	38.15 ± 1.14
	15	–	–	–	2.70 ± 0.08	36.63 ± 1.10
	20	287 ± 5	368 ± 5	81	2.75 ± 0.08	35.60 ± 1.10
	25	294 ± 5	439 ± 5	145	2.85 ± 0.09	34.00 ± 1.02
	30	306 ± 5	456 ± 5	150	2.93 ± 0.09	32.70 ± 1.00
	33	314 ± 5	446 ± 5	132	2.98 ± 0.09	32.00 ± 1.00
$(50-x)Na_2O–xZnO–50P_2O_5$	5	283 ± 5	–	–	2.51 ± 0.07	41.00 ± 1.30
	10	287 ± 5	310 ± 5	23	2.53 ± 0.08	41.00 ± 1.23
	15	290 ± 5	–	–	2.65 ± 0.08	39.60 ± 1.23
	20	295 ± 5	320 ± 5	25	2.68 ± 0.08	39.54 ± 1.20
	25	296 ± 5	330 ± 5	34	2.71 ± 0.08	39.50 ± 1.20
	30	305 ± 5	360 ± 5	55	2.78 ± 0.08	38.80 ± 1.20
	33	311 ± 5	400 ± 5	89	2.80 ± 0.09	38.74 ± 1.20

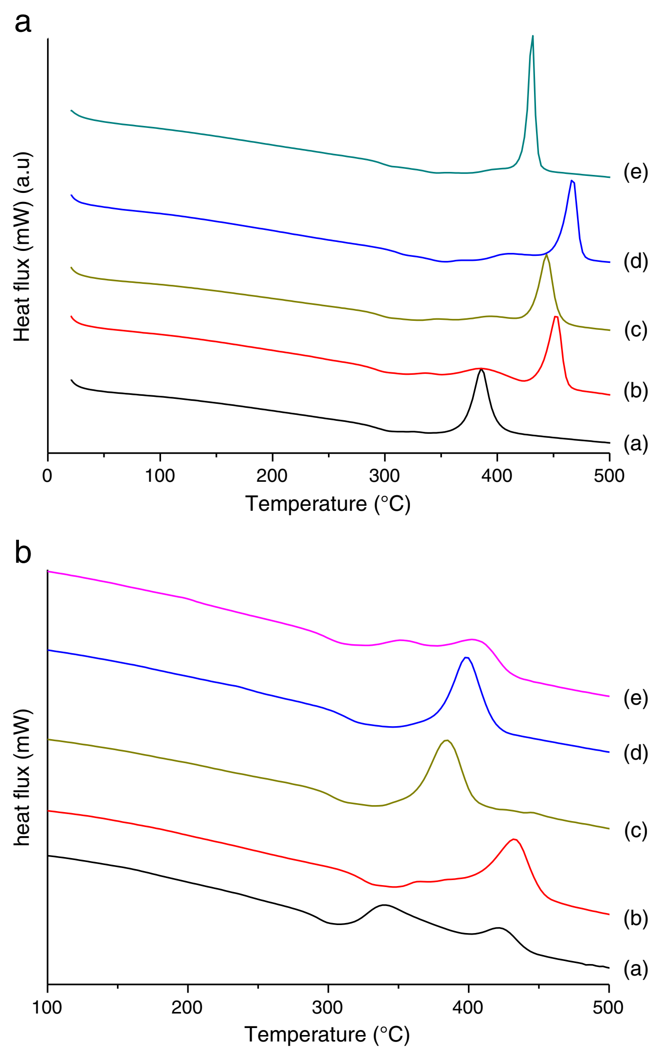

Fig. 3. a. DSC curves of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ glasses $(0 \leq x \leq 33$ mol%) over ZnO content (a)10 mol% ZnO, (b)20 mol% ZnO, (c)25 mol% ZnO, (d)30 mol% ZnO, (e)33 mol% ZnO. b. DSC curves of $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ glasses $(0 \leq x \leq 33$ mol%) over ZnO content (a)10 mol% ZnO, (b)20 mol% ZnO, (c)25 mol% ZnO, (d)30 mol% ZnO, (e)33 mol% ZnO.

introduced (Fig. 4). A similar evolution of glass transition temperature has been obtained for the same glass compositions of the first series by Montagne et al. [17].

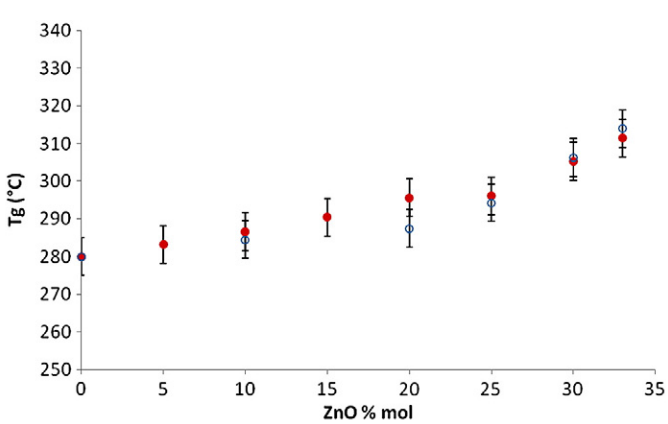

Fig. 4. T_g dependence of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ (○)and $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ (●) $(0 \leq x \leq 33$ mol%) glasses over ZnO content.

This result is probably due to the incorporation of metallic ion (Zn^{2+}) with a higher charge which serves to increase the atomic packing and strengthen the glass network.

On the other hand, the main reason for the increase in T_g values is related to the bond strength of the glass network due to the formation of P–O–Zn ionic bond when ZnO oxide is added.

A similar behavior was observed for other phosphate glasses containing Cd^{2+} and Pb^{2+}. Glass transition temperature increases from 292 °C to 419 °C in $(40-x)$–Na_2O–$xCdO$–$10PbO$–$50P_2O_5$ $(0 \leq x \leq 40$ mol%) [9], and from 282 °C to 378 °C in $(50-x)$–Na_2O–$xPbO$–$50P_2O_5$ $(0 \leq x \leq 40$ mol%) [24].

According to Dietezel, the thermal stability of glasses (ΔT) can be expressed by the temperature difference between T_g and T_c, $\Delta T = T_c - T_g$, where T_g and T_c are the glass transition and crystallization temperature as mentioned in Table 2 [25]. Increasing ΔT delays the nucleation process, indicating a better stability of the glass [25]. This result can also be explained by the electric field strength (z/r^2) which is larger for Zn^{2+} than Na^+ [9].

The nature of bonding is responsible for the variation of T_g. Na_2O rich glasses are predominately ionic, and become covalent with a higher level of ZnO, leading to the formation of P–O–Zn linkage in the vitreous network which ameliorates their chemical durability [16,17].

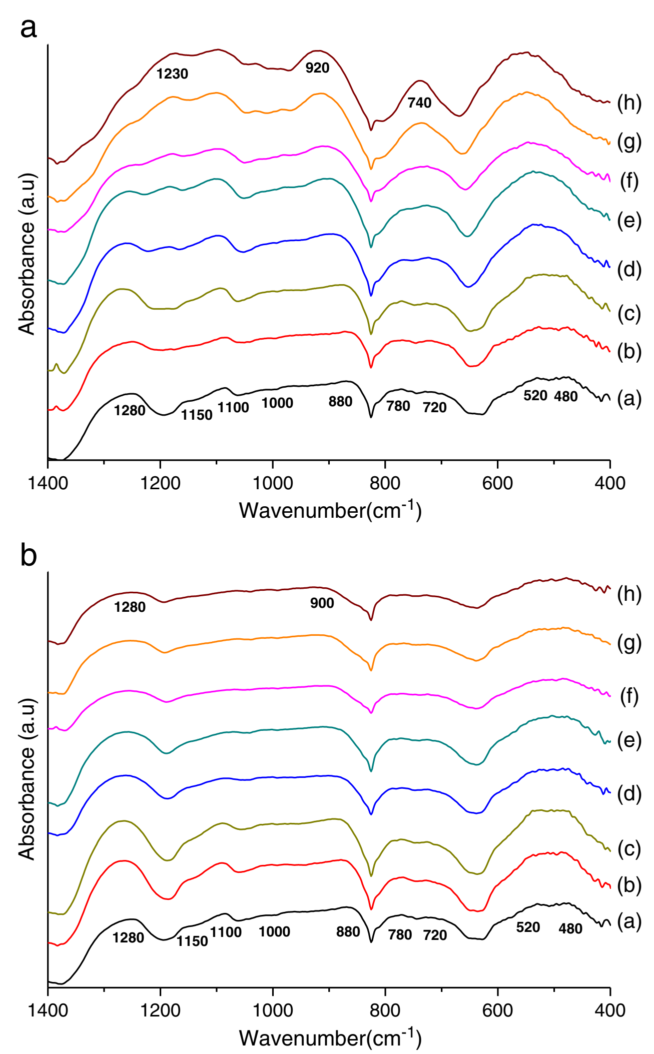

Fig. 5. a. Infrared spectra of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ glasses: (a)0 mol% ZnO, (b)5 mol% ZnO, (c)10 mol% ZnO, (d)15 mol% ZnO, (e)20 mol% ZnO, (f)25 mol% ZnO, (g)30 mol% ZnO, (h)33 mol% ZnO. b. Infrared spectra of $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ glasses: (a)0 mol% ZnO, (b)5 mol% ZnO, (c)10 mol% ZnO, (d)15 mol% ZnO, (e)20 mol% ZnO, (f)25 mol% ZnO, (g)30 mol% ZnO, (h)33 mol% ZnO.

Table 3
Infrared and Raman band assignments (cm^{-1}) of $(50 - x/2)Na_2O–xZnO–(50 - x/2)P_2O_5$ ($0 \leq x \leq 33$ mol%).

x	ν_{as} (PO_2)		ν_s (PO_2)		ν_{as} (PO_3)		ν_s (PO_3)		ν_{as} (POP)		ν_s (POP)		(PO_4^{3-})	
	IR	Raman	IR	Raman	IR	Raman	IR	Raman	IR	Raman	IR	Raman	IR	Raman
0	1280vw	1274vw	1150vw	1164vs	1100vw	–	1000vw	–	880vw	–	780–720vw	685 m	535–480w	380vw
5	1280vw	–	–	–	–	–	–	–	880vw	–	780–720vw	–	535–480w	–
10	1280vw	–	–	1164vs	–	1020vw	–	–	880vw	–	780–720vw	685 m	535–480w	380vw
15	1280vw	–	–	–	–	–	–	–	880vw	–	780–720vw	–	535–480w	–
20	1265vw	–	–	1164 m	–	1020w	–	–	880vw	–	780–720vw	685w	535–480w	380vw
25	1245vw	–	–	1164w	–	1020 m	–	–	900vw	–	780–720w	685w	535–480w	380vw
30	1250vw	–	–	1164w	–	1020 s	–	–	920 m	–	740 m	685 m	545vw	380vw
33	1240vw	–	–	1164vw	–	1020vs	–	–	920 m	–	740 m	685vw	550vw	380vw

3.3. Infrared and Raman spectroscopies

FTIR spectra of $(50 - x)Na_2O–xZnO–(50 - x/2)P_2O_5$ and $(50 - x)Na_2O–xZnO–50P_2O_5$ glasses are shown in Fig. 5 (a, b). Infrared and Raman band assignments are listed in Tables 3 and 4.

The characteristic features of metaphosphate glasses $NaPO_3$ ($x = 0$) are: the PO_2 asymmetric stretching vibration band ($\upsilon_{as}(PO_2)$) near 1280 cm^{-1}, the PO_2 symmetric stretching vibration band ($\upsilon_s(PO_2)$) at 1150 cm^{-1}, the $\upsilon_{as}(PO_3)$ groups (chain end groups) at 1100 cm^{-1}, the υ_s of PO_3 groups at 1000 cm^{-1}, the υ_{as} of P–O–P groups at 880 cm^{-1}, the υ_s of P–O–P groups at 780 and 720 cm^{-1} and the deformation mode of $P–O^-(PO_4^{3-})$ groups at 535 and 480 cm^{-1} [5,9,10,14,16,26].

As ZnO is introduced, the asymmetric band of PO_2 shifts from 1280 cm^{-1} to 1230 cm^{-1} in $33.5Na_2O–33ZnO–33.5P_2O_5$ glass indicating the depolymerization of phosphate chains when x increases [9].

The FTIR spectra revealed that the asymmetric band of P–O–P stretching mode shifted from 880 cm^{-1} for $NaPO_3$ to 920 cm^{-1} in $33.5Na_2O–33ZnO–33.5P_2O_5$ glass. This result can be explained by the increase of covalent character of P–O–P bonds that strengthened as the Na_2O is replaced by divalent cation (such as Zn^{2+}) [9,18]. It may be also attributed to the shortening of phosphate chain length due to the higher field strength and the size of the metallic cation [5,9,10,14], when the ratio O/P increases for $(50 - x/2)Na_2O–xZnO–(50 - x/2)P_2O_5$. This behavior is similar to copper, cadmium and lead phosphate glasses reported in literature [5,9].

The FTIR spectra of $NaPO_3$ glasses revealed also two bands in the frequency range of 780–720 cm^{-1} which are attributed to the presence of two P–O–P bridges in metaphosphate chains based on $(P_2O_6)^{2-}$ groups (Fig. 5 (a, b)) [10,14]. However, it's interesting to note that for the first series glasses containing 30 and 33 mol% ZnO exhibit only a single band at 740 cm^{-1} assigned to the P–O–P linkage in pyrophosphate group $(P_2O_7)^{4-}$ (Fig. 5a) [10,14]. These changes depend on the composition of glasses. For the first series, as the content of ZnO increases, the infinite chain metaphosphate glasses are gradually broken leading to the depolymerization of the skeleton of $(P_2O_6^{2-})_\infty$ into short phosphate groups such as: $P_2O_7^{4-}$ and PO_4^{3-} [14].

The Raman spectra of $(50 - x/2)Na_2O–xZnO–(50 - x/2)P_2O_5$ glasses, where O/P varies, with $0 \leq x \leq 33$ mol% ZnO are shown in Fig. 6a. The Raman spectrum of $NaPO_3$ (a) reveals broad band at about 1274 cm^{-1} and three sharp peaks at about 380, 685 and 1164 cm^{-1}. The bands situated at 1274 and 1164 cm^{-1} are assigned to the asymmetric and symmetric vibrations ($\nu_{as}(PO_2)$ and $\nu_s(PO_2)$) of the non-bridging oxygen atoms (NBO) bonded to phosphorus atoms (O–P–O) in metaphosphate chains (Q^2) [10,14,26,27]. The wide band at 685 cm^{-1} is assigned to the symmetric vibration of the bridging oxygen connecting two PO_4 tetrahedron (P–O–P) (Q^2) in phosphate chains [10,14,27]. The low frequency attributed to the faint band at 380 cm^{-1} is related to the bending motion of phosphate polyhedral [10].

For the first series of glasses, a new bond appeared for x = 10%mol at about 1020 cm^{-1} attributed to the symmetric stretching vibration $\nu_s(PO_3)$ of the NBO against the P atoms in PO_4 tetrahedron with 3 NBO in pyrophosphate groups (Q^1) [27].

With increasing ZnO content, we observe some decrease of the overall background in the ranges of 600–800 cm^{-1} and 1100–1300 cm^{-1} [10,27]. These changes can be attributed to the disruption of P–O–P bridges and the depolymerization of phosphate chains. As a result, the formation of phosphate dimers (Q^1) when the O/P ratio increases.

From Fig. 6a, one can note that the intensity of bands situated at about 1164 and 685 cm^{-1} decreases with increasing ZnO content. In addition, the Raman spectra revealed the displacement of these bands to 1180(d, e) and 780 cm^{-1} (d, e, f) with 30 and 33 mol% of ZnO concentration. This result can be explained by the high P–NBO π character due to the low condensation of phosphate chains when ZnO oxide is gradually incorporated.

For the second series of glasses (O/P = 3), one can noticed no change in the spectra for all glass composition investigated by Raman and infrared spectroscopies. These results may be due to the conservation of structure which is essentially based on metaphosphate groups (Q^2) as shown in Figs. 5b and 6b.

3.4. ^{31}P MAS-NMR spectroscopy

The ^{31}P MAS-NMR spectra of both series of glasses are shown in Fig. 7 (a, b). The remaining peaks observed on both sides of the centerbands represent the spinning sidebands that arise from the

Table 4
Infrared and Raman band assignments (cm^{-1}) of $(50 - x)Na_2O–xZnO–50P_2O_5$ ($0 \leq x \leq 33$ mol%).

x	ν_{as} (PO_2)		ν_s (PO_2)		ν_{as} (PO_3)		ν_s (PO_3)		ν_{as} (POP)		ν_s (POP)		(PO_4^{3-})	
	IR	Raman	IR	Raman	IR	Raman	IR	Raman	IR	Raman	IR	Raman	IR	Raman
0	1280vw	1274vw	1150vw	1164vs	1100vw	–	1000vw	–	880vw	–	780–720vw	685 m	535–480w	380vw
5	1280vw	1274vw	–	1164vs	1100vw	–	–	–	880vw	–	780–720vw	685 m	535–480w	380vw
10	1280vw	–	–	1164vs	1100vw	–	–	–	880vw	–	780–720vw	685 m	535–480w	380vw
15	1280vw	–	–	1164 m	–	–	–	–	880vw	–	780–720vw	685 m	535–480w	380vw
20	1280vw	–	–	1164 m	–	–	–	–	880vw	–	780–720vw	685w	535–480w	380vw
25	1280vw	–	–	1164 m	–	–	–	–	900vw	–	780–720w	685w	535–480w	380vw
30	1280vw	–	–	1164 m	–	–	–	–	900 m	–	780–720vw	685 m	535–480vw	380vw
33	1280vw	–	–	1164 m	–	–	–	–	900 m	–	780–720vw	685 m	535–480vw	380vw

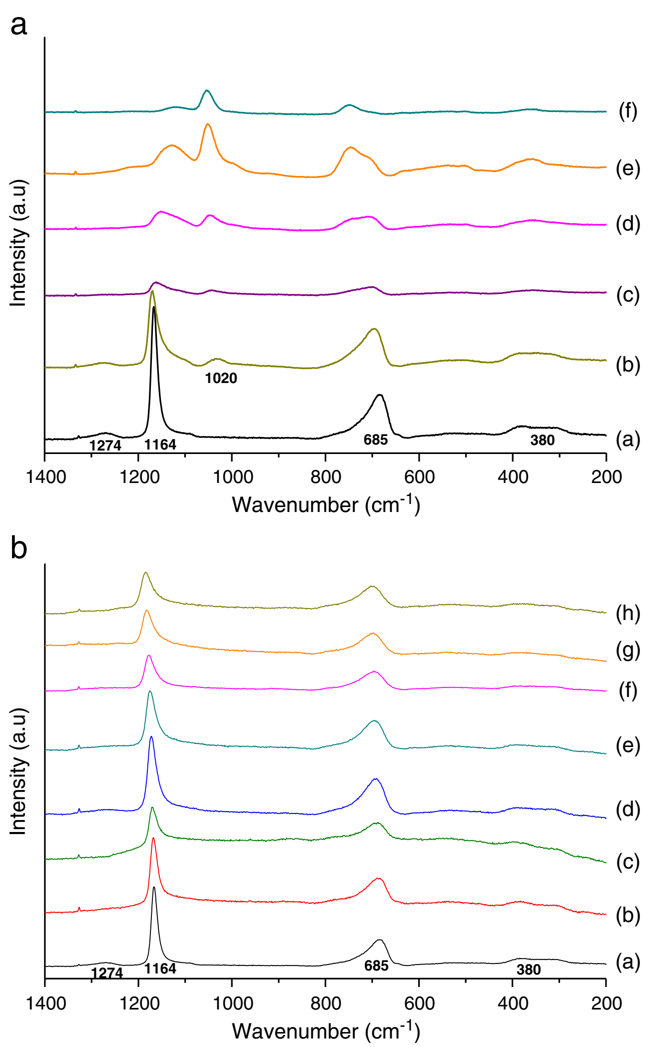

Fig. 6. a. Raman spectra of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ glasses: (a)0 mol% ZnO, (b)10 mol% ZnO, (c)20 mol% ZnO, (d)25 mol% ZnO, (e)30 mol% ZnO, (f)33 mol% ZnO. b. Raman spectra of $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ glasses: (a)0 mol% ZnO, (b)5 mol% ZnO, (c)10 mol% ZnO, (d)15 mol% ZnO, (e)20 mol% ZnO, (f)25 mol% ZnO, (g)30 mol% ZnO, (h)33 mol% ZnO.

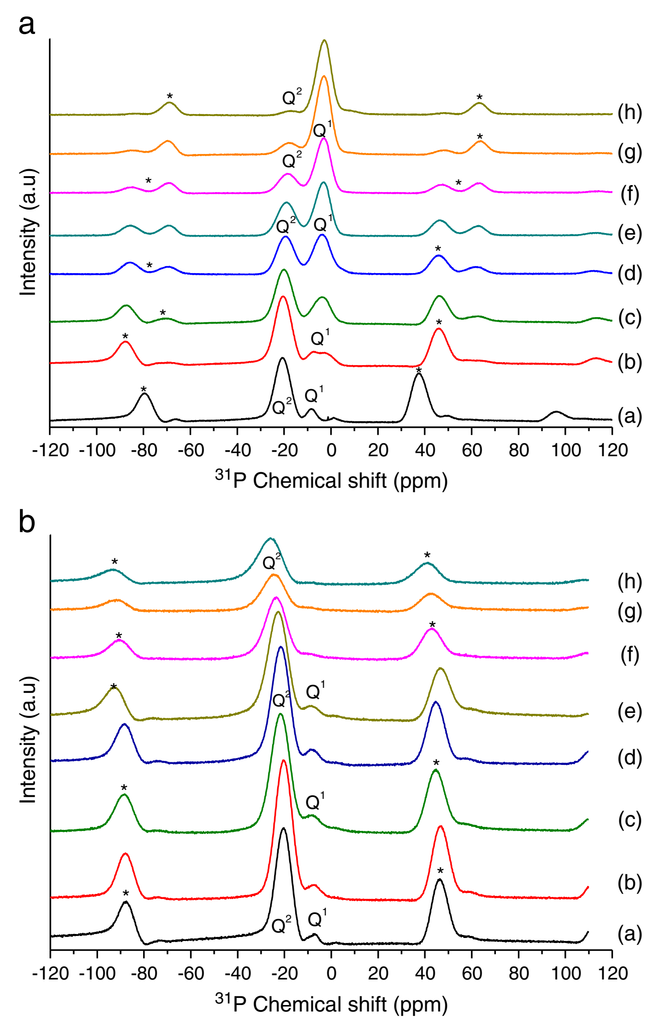

Fig. 7. a. ^{31}P MAS-NMR spectra of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ glasses: (a)0 mol% ZnO, (b)5 mol% ZnO, (c)10 mol% ZnO, (d)15 mol% ZnO, (e)20 mol% ZnO, (f)25 mol% ZnO, (g)30 mol% ZnO, (h)33 mol% ZnO. The spinning sidebands are marked with asterisks. b. ^{31}P MAS-NMR spectra of $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ glasses: (a)0 mol% ZnO, (b)5 mol% ZnO, (c)10 mol% ZnO, (d)15 mol% ZnO, (e)20 mol% ZnO, (f)25 mol% ZnO, (g)30 mol% ZnO, (h)33 mol% ZnO. The spinning sidebands are marked with asterisks.

chemical shift anisotropy interaction associated with these units [1]. Isotopic peaks are labeled with their respective chemical shift of Q^n units. The major feature of $NaPO_3$ glass is the existence of an isotropic peak at −21.0 ppm representing the Q^2 tetrahedral sites that are the bases of metaphosphate chains [5,18]. The peak at −6.9 ppm is assigned to Q^1 sites (chain end groups) [18].

Referring to the literature, some authors have reported the chemical shift at +1.4 ppm for chain end groups of $NaPO_3$ [5,18]. The major feature of the spectra is the existence of the two isotropic peaks in the ranges near −21.0 ppm to −18.8 ppm and −6.9 to −3.9 ppm for the first series. As ZnO is added to metaphosphate chains, the intensity of peak attributed to Q^1 tetrahedral sites increases with ZnO content and becomes the dominant spectral feature in $33.5Na_2O$–$33ZnO$–$33.5P_2O_5$ glass. Whereas, the other isotropic peaks assigned to Q^2 tetrahedral sites decrease with increasing ZnO content [5,18]. This result is in good agreement with the structural study of $(100-x)NaPO_3$–$xZnO$ glasses ($0 \le x \le 33.3$ mol%) performed by Montagne et al. [18].

Fig. 6a indicates that the phosphate glasses with 33 mol% of ZnO have almost only the Q^1 tetrahedral sites assigned to pyrophosphate groups which suggest that phosphate chains are totally depolymerized. On the other hand, the phosphorus chemical shift depends essentially on the phosphorus–ligand bond (P–O) for phosphate compounds and the electronic density of the non-bridging oxygens (NBO) [18]. Tables 5 and 6 report the average values of the measured chemical shifts for Q^1 and Q^2 sites. It mentioned that the Q^2 chemical shift becomes less shielded when ZnO is added. This decrease is probably due to the higher electronegativity of Zn^{2+} compared to Na^+ also the increase of π fraction of P-NBO resulting from the decondensation of phosphate chains

Table 5
Chemical shifts (ppm relative to 85% H_3PO_4) and relative concentration of the Q^n units determined by ^{31}P MAS-NMR and distribution of different Q species (%) in $(50-x/2)$ Na_2O–$xZnO$–$(50-x/2)P_2O_5$ glasses from their chemical composition (Cc) (with $C(Q^1) = 1 - C(Q^2)$).

X	Q^2(ppm) ±0.2	Q^2 fraction ± 0.02	$C(Q^2)$	Q^1(ppm) ±0.2	Q^1 fraction ±0.02	$C(Q^1)$	L_{CH}
0	−21.0	0.90	1	−6.9/+1.4	0.10	0	∞
5	−21.1	0.83	0.89	−5.21	0.17	0.11	19
10	−21.0	0.71	0.78	−4.6	0.29	0.12	9
15	−20.8	0.48	0.65	−4.6	0.52	0.35	6
20	−20.1	0.36	0.5	−3.8	0.64	0.5	4
25	−19.7	0.24	0.33	−4.2	0.76	0.67	3
30	−18.8	0.11	0.14	−4.0	0.89	0.86	2
33	−18.8	0.02	0.01	−3.9	0.98	0.99	2

Table 6
Chemical shifts (ppm relative to 85% H_3PO_4) and relative concentration of the Q^n units determined by ^{31}P MAS-NMR and distribution of different Q species (%) in $(50-x)$ Na_2O–$xZnO$–$50P_2O_5$ glasses from their chemical composition (Cc) (with $C(Q^1) = 1 - C(Q^2)$).

X	Q^2(ppm) ±0.2	Q^2 fraction ±0.02	$C(Q^2)$	Q^1(ppm) ±0.2	Q^1 fraction ±0.02	$C(Q^1)$	L_{CH}
0	−21.0	0.90	1	−6.9/+1.4	0.10	0	∞
5	−21.0	0.80	1	−6.8	0.20	0	∞
10	−22.0	0.96	1	−7.7	0.04	0	∞
15	−22.2	0.98	1	−7.9	0.02	0	∞
20	−22.8	0.97	1	−8.8	0.03	0	∞
25	−23.5	1.00	1	–	0	0	∞
30	−24.7	1.00	1	–	0	0.	∞
33	−26.6	1.00	1	–	0	0	∞

when ZnO is incorporated [18]. This suggests that Zn^{2+} ions are only bonded to pyrophosphate groups described by Q^1 tetrahedral sites [18]. As a result the increase of shielding Q^2 sites from −21.0 ppm in $NaPO_3$ glass to −18.8 ppm in $33.5Na_2O$–$33ZnO$–$33.5P_2O_5$.

For the second series of glasses, $(50-x)Na_2O$–$xZnO$–$50P_2O_5$, ^{31}P MAS-NMR spectra show the existence of two isotropic peaks, the first one is situated in the ranges near −6.9 ppm to −8.8 ppm and vanished for $x = 25$ mol%. The second peak is accentuated at −21.0 ppm for $NaPO_3$ to −26.6 ppm for $17Na_2O$–$33ZnO$–$50P_2O_5$ glass composition.

Fig. 7b indicates that for the higher ZnO content, ^{31}P MAS-NMR spectra shows the existence only the Q^2 tetrahedral sites assigned to metaphosphate groups suggesting that phosphate chains are not modified with the increase of ZnO concentration.

For the first series of glasses, the average values of the measured chemical shifts for Q^1 and Q^2 sites are reported in Table 5. This latter, mentioned that the Q^2 chemical shift becomes more shielded when ZnO content rises with a constant ratio O/P. This variation is probably due to the increase of the cross-link strength and the rigidity of the vitreous structure.

The effect of ZnO oxide can be quantified by determining the variation of the amount of Q^n species over the composition. ^{31}P MAS NMR is a suitable method to evaluate the Q species distributions in phosphate glasses.

However, it is possible to estimate the modification resulting from the incorporation of ZnO oxide in the vitreous network by calculating the amount of the different modes of connection of PO_4 tetrahedra, Q^2 and Q^1, on the base of their chemical composition, using the formula proposed by Van Wazer as shown in Tables 5 and 6 [27]:

$$C\left(Q^2\right) = 100^*(2-R) \quad \text{(a)}$$

where $R = ([ZnO] + [Na_2O])/[P_2O_5]$, and [ZnO], $[Na_2O]$, $[P_2O_5]$ are the %mol of the corresponding oxides [27].

The average chain length, $\mathbf{L_{CH}}$, defined as the number of P atoms in the chain can also be calculated by an expression proposed by Van Wazer [27]:

$$(1/\mathbf{L_{CH}}) = (R-1)/2 \quad \text{(b)}$$

where R is the molar ratio defined in Eq. (a).

Calculations are listed in Tables 5 and 6. These results show that polyphosphate chains became shorter as ZnO oxide is introduced for the first series of glasses and they are conserved for the second one.

The relative concentrations of the Q^n units obtained by integrating the peak surface obtained from the solid state NMR analysis of full spectra are mentioned in Tables 5 and 6. The total area of each approximated isotropic peak attributed to Q^2 and Q^1 tetrahedral sites was calculated in order to get information about their relative fraction intensity. The precision in the relative concentrations of the Q^n sites is estimated to ±2%.

For the first series of glasses, Table 5 indicates that metaphosphate chains became shorter as ZnO oxide is gradually incorporated as a result

Table 7
Molar dissolution heat of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ and $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ ($0 \le x \le 33$ mol%)glasses.

	X	$\Delta_{sol}H_{(ZnO)}$ (kJ mol^{-1})
$(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$	0	4.80 ± 0.40
	5	3.70 ± 0.20
	10	2.92 ± 0.15
	15	1.70 ± 0.10
	20	−1.15 ± 0.10
	25	−3.21 ± 0.2
	30	−6.34 ± 0.32
	33	−13.05 ± 1.00
$(50-x/)Na_2O$–$xZnO$–$50P_2O_5$	0	4.80 ± 0.40
	5	3.65 ± 0.30
	10	3.02 ± 0.22
	15	2.40 ± 0.20
	20	1.94 ± 0.14
	25	0.92 ± 0.1
	30	0.45 ± 0.03
	33	0.04 ± 0.00

the depolymerization of the phosphate chains which can be quantitatively described by the Q^n distributions.

For $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ series of glasses, the concentration of Q^2 is prominent for all glass composition when their structure is described by an infinite polyphosphate chains, as shown in Table 6. These results are in good agreement with those predicted theoretically from their chemical composition.

3.5. Calorimetric study of glasses

As mentioned previously, the calorimetric study was performed by determining the energy resulting from the dissolution of the glasses in a suitable solvent.

Phosphate glasses are soluble in mineral acids [20,21]. However, for the sake of comparison, the dissolution should be complete and give rise to any secondary phenomena. Preliminary experiments performed by recording the calorimetric profile of dissolution in various solvents showed that the 4.5% weight of phosphoric acid solution is the most suitable solvent (such as: HNO_3, HCl, NaOH, KOH, CH_3COOH).

Experiments were carried out by dissolving the same mass of solids (20 mg) in 4.5 ml of solvent. Table 7 summarizes the heat dissolution of both series of glasses.

Complementary experiments were realized for both series of glasses in order to determine the relative error on the integrated area using a statistical calculation method developed by Pattengill et al. [28].

The plots of heat dissolution of glasses ($\Delta_{sol}H$ (kJ mol^{-1})) are shown in Fig. 8. For the first series, one can notice that the dissolution phenomenon is endothermic for the low ZnO composition glasses and becomes exothermic when ZnO content increases. It seems that the change in thermal signs is related to the progressive depolymerization of

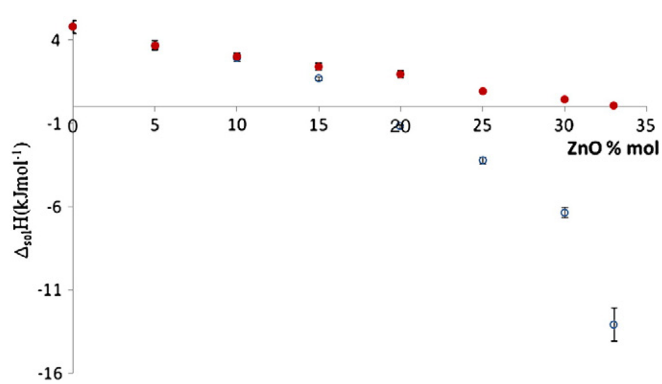

Fig. 8. Evolution of heat dissolution of $(50-x/2)Na_2O$–$xZnO$–$(50-x/2)P_2O_5$ (○) and $(50-x)Na_2O$–$xZnO$–$50P_2O_5$ (●) ($0 \le x \le 33$ mol%) glasses.

metaphosphate groups (Q^2) in order to give pyrophosphate units (Q^1) as ZnO is incorporated in the structure.

For the second series of glasses where O/P = 3, the calorimetric dissolution exhibits only endothermic effect for all of the composition domain ($0 \leq x \leq 33$ mol%). We observed a monotonically diminution of $\Delta_{sol}H$ values, which is probably due to the replacement of Na^+ by Zn^{2+}.

Spectroscopic investigations such as FTIR, Raman and ^{31}P MAS-NMR revealed the formation of pyrophosphate groups for $(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ ($3 \leq O/P \leq 3.49$) glass composition, resulting from the cleavage of the P–O–P bridges when the amount of ZnO oxide is progressively increases. This result is in good correlation with the thermochemical study, density measurements, molar volume evolutions and glass transition temperature variations which suggest the compactness of vitreous structure.

For $(50 - x)Na_2O$–$xZnO$–$50P_2O_5$ (O/P = 3) glass compositions, there is no structural change occurred when increasing ZnO concentration.

4. Conclusion

Glasses of $(50 - x/2)Na_2O$–$xZnO$–$(50 - x/2)P_2O_5$ and $(50 - x)Na_2O$–$xZnO$–$50P_2O_5$ ($0 \leq x \leq 33$ mol%) compositions have been investigated using FTIR, Raman and MAS-NMR spectroscopies in order to study the structural role of ZnO oxide for both series of glasses.

For the first glass series ($3 \leq O/P \leq 3.49$), ZnO oxide breaks the infinite metaphosphate chains $(P_2O_6)_\infty^{2-}$ into smaller groups such as: $P_2O_7^{4-}$ suggesting the depolymerization of phosphate chains. This result can be correlated to the change of calorimetric dissolution phenomenon in 4.5% weight of H_3PO_4 solution.

Contrary to the first series, for the second one (O/P = 3), structural investigations show the conservation of linear chains (metaphosphate groups) which can be related to the monotonic variation of the endothermic dissolution phenomenon over all the glass composition range.

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